

MARKING SCHEME

NETWORK DESIGN CAT1 TERM 3

Question 1 – 4 Marks

a) Definition of a computer network – 2 marks

- A computer network is a collection of two or more interconnected computers/devices that communicate and share resources/data through communication media. **(2 marks)**

b) Benefits of computer networking – 2 marks

Award **1 mark each** for any two:

- Resource sharing, e.g. printers and storage.
- File/data sharing.
- Communication between users.
- Sharing of internet connection.
- Centralized management.
- Reduced operational costs.
- Improved collaboration.

Question 2 – 8 Marks

Award **2 marks for each correctly explained network type.**

a) LAN – 2 marks

- Local Area Network. **(1 mark)**
- Covers a small geographical area such as a room, office, building, school or laboratory. **(1 mark)**

b) WAN – 2 marks

- Wide Area Network. **(1 mark)**
- Covers a large geographical area such as cities, countries or continents. **(1 mark)**

c) MAN – 2 marks

- Metropolitan Area Network. **(1 mark)**

- Covers a city or metropolitan area and is larger than a LAN but smaller than a WAN. **(1 mark)**

d) PAN – 2 marks

- Personal Area Network. **(1 mark)**
- Covers a very small area around an individual, connecting devices such as phones, laptops and Bluetooth devices. **(1 mark)**

Total: 8 marks

Question 3 – 10 Marks

Award **2 marks for each topology.**

a) Star topology – 2 marks

- All network devices are connected to a central device such as a switch or hub. **(1 mark)**
- Data communication between devices normally passes through the central device. **(1 mark)**

b) Ring topology – 2 marks

- Devices are connected in a closed loop/ring. **(1 mark)**
- Data travels from one device to another around the ring. **(1 mark)**

c) Mesh topology – 2 marks

- Devices are interconnected through multiple links. **(1 mark)**
- Provides alternative paths for data, improving reliability. **(1 mark)**

d) Hybrid topology – 2 marks

- Combines two or more different network topologies. **(1 mark)**
- Example: a network combining star and bus topologies. **(1 mark)**

e) Point-to-point topology – 2 marks

- Provides a direct connection between two devices. **(1 mark)**
- The connection is dedicated to communication between the two devices. **(1 mark)**

Total: 10 marks

Question 4 – 8 Marks

Award 2 marks for each correctly stated and explained component.

Accept any four of the following:

a) Switch

- Connects devices within a network. **(1 mark)**
- Forwards data to the appropriate destination device using MAC addresses. **(1 mark)**

b) Router

- Connects different networks. **(1 mark)**
- Forwards packets between networks using IP addresses/routing information. **(1 mark)**

c) Computers/End devices

- Devices used by network users to access network resources and services. **(1 mark)**
- Examples include desktop computers, laptops and servers. **(1 mark)**

d) Network ports

- Interfaces used to connect network devices or cables. **(1 mark)**
- Enable transmission and reception of network data. **(1 mark)**

e) Transmission media

- Provides the path through which data travels between network devices. **(1 mark)**
- Examples include UTP, fibre optic cable, coaxial cable and wireless media. **(1 mark)**

Total: 8 marks

Question 5 – 6 Marks

Award 2 marks for each correctly stated medium and characteristic.

Any three:

a) UTP cable

- Unshielded Twisted Pair cable. **(1 mark)**
- Consists of twisted pairs of copper wires and is commonly used in Ethernet networks. **(1 mark)**

b) Fibre optic cable

- Transmits data using light through optical fibres. **(1 mark)**
- Provides high bandwidth and is resistant to electromagnetic interference. **(1 mark)**

c) Coaxial cable

- Consists of a central conductor surrounded by insulation and shielding. **(1 mark)**
- Provides protection against electromagnetic interference. **(1 mark)**

d) Wireless

- Uses radio waves to transmit data without physical cables. **(1 mark)**
- Allows mobile/flexible connectivity. **(1 mark)**

Total: 6 marks

Question 6 – 6 Marks

Award 2 marks for each stage.

a) User requirements identification – 2 marks

- Determining what users need from the proposed network. **(1 mark)**
- Involves collecting information such as number of users, devices, applications and required services. **(1 mark)**

b) User requirements analysis – 2 marks

- Examining and evaluating the collected requirements. **(1 mark)**
- Determines network capacity, performance, security, budget and other technical requirements. **(1 mark)**

c) User requirements documentation – 2 marks

- Recording the identified and analysed requirements in a clear and organized form. **(1 mark)**
- Provides a reference for designing, implementing and evaluating the network. **(1 mark)**

Total: 6 marks

Question 7 – 4 Marks

Physical network design – 2 marks

- Describes the physical arrangement of network devices, cables, racks, access points and other hardware. **(1 mark)**
- Shows where the network components will be physically installed. **(1 mark)**

Logical network design – 2 marks

- Describes how data and network services will logically flow through the network. **(1 mark)**
- Includes aspects such as IP addressing, subnets, VLANs, routing and logical connectivity. **(1 mark)**

Question 8 – 2 Marks

Greenfield network site – 1 mark

- A new site where the network is being designed and installed without significant existing network infrastructure. **(1 mark)**

Brownfield network site – 1 mark

- An existing site with network infrastructure where the new network must work with, modify or upgrade the existing infrastructure. **(1 mark)**

Question 9 – 2 Marks

Award **1 mark each** for any two correctly stated fundamental network design goals:

- **Scalability** – ability of the network to accommodate growth and additional users/devices.
- **Availability** – ability of the network to remain operational and accessible when required.
- **Security** – protection of network resources and data from unauthorized access and threats.
- **Manageability** – ease of monitoring, maintaining, configuring and troubleshooting the network.

Total: 2 marks